

Question	Answer	Marks
2(a)	force per unit positive charge	B1
2(b)(i)	four parallel lines perpendicular to the plates, each starting on one plate and finishing on the other	B1
	four parallel lines, approximately evenly distributed over the region between the plates	B1
	arrows showing direction as upwards	B1

Question	Answer	Marks
2(b)(ii)	gravitational force is downwards so electric force must be upwards	B1
	electric force is upwards so Q must be positive	B1
2(c)(i)	$E = V / x$	A1
2(c)(ii)	$W = VQ / x$	A1
2(d)	$W = mg$ <u>and</u> $g = 9.81$	C1
	mass = $(4800 \times 3 \times 1.60 \times 10^{-19}) / (0.024 \times 9.81)$ = $9.8 \times 10^{-15} \text{ kg}$	A1
2(e)	both (electric and gravitational) forces are now downwards	B1
	droplet will <u>accelerate</u> downwards	B1